Bit	7	6	5	4	3	2	1	0
	FOC0	WGM00	COM01	COM00	WGM01	CS02	CS01	CS00
Read/Write	W	RW	RW	RW	RW	RW	RW	RW
Initial Value	0	0	0	0	0	0	0	0

FOC0 D7 Force Output compare: Writing 1 to it forces the wave generator to act as if a compare match has occurred.

WGM01:00 D6 D3 Timer0 mode selector bits

0	0	Normal
0	1	CTC (Clear Timer on Compare match)
1	0	PWM, phase correct
1	1	Fast PWM

COM01:00 D5 D4 Compare Output Mode; The table shows what the wave generator does on compare match when the timer is in Normal or CTC mode:

COM01	COM00	Description
0	0	Normal port operation, OC0 disconnected
0	1	Toggle OC0 on compare match
1	0	Clear OC0 on compare match
1	1	Set OC0 on compare match

CS02:00 D2 D1 D0 Timer0 clock selector

0	0	0	No clock source (Timer/Counter stopped)
0	0	1	clk (no prescaling)
0	1	0	clk / 8
0	1	1	clk / 64
1	0	0	clk / 256
1	0	1	clk / 1024
1	1	0	External clock source on T0 pin. Clock on falling edge
1	1	1	External clock source on T0 pin. Clock on rising edge

Figure 15-4. TCCR0 (Timer/Counter Control Register) Register

Bit	7	6	5	4	3	2	1	0
	OCF2	TOV2	ICF1	OCF1A	OCF1B	TOV1	OCF0	TOV0
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

TOV0 D0 Timer0 overflow flag bit
0 = Timer0 did not overflow.
1 = Timer0 has overflowed (going from $FF to $00).

OCF0 D1 Timer0 output compare flag bit
0 = compare match did not occur.
1 = compare match occurred.

TOV1 D2 Timer1 overflow flag bit

OCF1B D3 Timer1 output compare B match flag

OCF1A D4 Timer1 output compare A match flag

ICF1 D5 Input Capture flag

TOV2 D6 Timer2 overflow flag

OCF2 D7 Timer2 output compare match flag

Figure 9-6. TIFR (Timer/Counter Interrupt Flag Register)

Write an assembly language program to generate a 750 Hz square wave on port A bit 1 (PA1) using Timer 0 with timer overflow. Use fosc = 16 MHz.

f = 750 Hz => T = 1/f = 1/750 = 0.0013333 s = 1333.33 x 10^-6 s = 1333.33 us

Thalf = 666.67 us = tdelay = n TMC => TMC = (1cc/1MC) (1s/16 (10^6) cc) = 0.0625 us/MC

tdelay = 666.67 us = n (0.0625 us/MC) => n = 10666.67 MC or 10667 MC

10667 > 255 so need timer prescaler

If use fosc/8, then TMC = (1 cc/1 MC)(8 (1s/16(10^6)cc)) = 0.5 us/MC => n = 1333 MC > 255 so can't work for 8-bit timer 0

If use fosc/64, then TMC = (1cc/1MC)(64 (1s/16(10^6)cc)) = 4 us/MC => n = 166.67 or 167 MCs < 256 so ok.

If use fosc/256, then TMC = (1cc/1MC)(256(1s/16(10^6)cc)) = 16 us/MC => n = 41.67 or 42 MCs < 256 so ok. Since each clock tick lasts for a longer duration as the fosc prescaler increases, the timing error is amplified. So, see to use the smallest fosc prescaler that works to mitigate the timing error. For this example, use fosc/64 (last 3 bits of TCCR0 are 011 (CS02:00).

ASSEMBLY PROGRAM

```
        .ORG 0
        SBI DDRA, 1; PA1 OUTPUT
        LDI R20, 0B00000010; SET BIT 1 FOR TOGGLING TO GENERATE WAVEFORM
        LDI R21, 0X00; INITIALIZE PA (IN PARTICULAR PA1 FOR WAVEFORM)
LOOP:   EOR R21, R20; TOGGLE BIT 1 FOR PA1 WAVEFORM
        OUT PORTA, R21; OUTPUT TOGGLED VALUE FOR PA1 FOR WAVEFORM
        CALL DELAY; CALL DELAY ROUTINE THAT USES TIMER 0 TIMER OVERFLOW BASED ON
                  ; CALCULATION ABOVE
        RJMP LOOP; SQUAVE WAVE GENERATION, IGNORING OVERHEAD FROM ADDITIONAL
        INSTRUCTIONS  IN HALF PERIOD TIME FOR WAVEFORM
DELAY:  LDI R18, -167; TIMER 0 PRELOAD VALUE BASED ON FOSC/64 PRESCALER
                  ; FOR 750 HZ HALF WAVE
        OUT TCNT0, R18
        LDI R18, 0XB00000011; NORMAL MODE, TOV0, FOSC/64 FOR TIMER 0 CONFIGURATION
        OUT TCCR0, R18
AGAIN:  IN R19, TIFR; WAIT FOR TOV0 TO BE SET
                  ; (TCNT0 TICKS UP TO 255 THEN TO 0, CAUSING TOV0 = 1
        SBRS, R18, TOV0; TOV0 = 1? OTHERWISE REPEAT
        RJMP AGAIN
        LDI R18, 0
```

```
        OUT TCCR0, R18; TURN OFF TIMER 0
        LDI R18, (1<<TOV0)
        OUT TIFR, R18; RESET TOV0 BY WRITING A 1 TO IT AND 0S TO THE
                       ; OTHER FLAG BITS OF TIFR
        RET
```

CONVERT THE ASSEMBLY PROGRAM TO C

```
#INCLUDE <AVR\IO.H>
VOID DELAYROUTINE();
INT MAIN()
{
        DDRA |= 0B00000010; // BIT 1 OUTPUT, OTHER BITS HAVE ORIGINAL STATUS
                            // (UNCHANGED)
        PORTA = 0X00;
        WHILE(1)
        {
                PORTA ^= 0B00000010; // TOGGLE BIT 1
                DELAYROUTINE();  // HALF PERIOD TIME DELAY
        }
        RETURN 0;
}
VOID DELAYROUTINE()
{
        TCNT0 = -167;  // TIMER 0 PRELOAD
        TCCR0 = 0B00000011; // TIMER 0 AS TIMER WITH PRESCALER FOSC/64, TURN ON TIMER 0
        WHILE((TIFR & (1<<TOV0)) == 0); // WAIT FOR TOV0 = 1 (TIMER 0 OVERFLOW)
        TCCR0 = 0;  // TURN OFF TIMER 0
        TIFR = 1<<TOV0;  // RESET TOV0 FLAG IN TIFR
}
```

Find the timer 0 delay in seconds from the program below. Exclude the overhead due to instructions in the loop. Fosc = 8 MHz

```
.ORG 0
LDI R20, HIGH(RAMEND)
OUT SPH, R20
LDI R20, LOW(RAMEND)
OUT SPL, R20
LDI R16, 0X10
SBI DDRB, 4
LDI R17, 0
OUT PORTB, R17
START:  RCALL DELAY
        EOR R17, R16
        OUT PORTB, R17
        RJMP START

; TIMER DELAY
DELAY:  LDI R20, 0X10
        OUT TCNT0, R20
        LDI R20, 0X03
        OUT TCCR0, R20
AGAIN:  IN R20, TIFR
        SBRS R20, TOV0
        RJMP AGAIN
        LDI R20, 0X00
        OUT TCCR0, R20
        LDI R20, 1<<TOV0
        OUT TIFR, R20
        RET
```

TCCR0 = 0000 0011

clk/64

TCNT0 = 0X10 = 16

256 − 16 = 240 ticks

$t_{delay} = (240)\left(\frac{64(1)}{8(10^6)}\right)$

$= 1920\,ms$

PB5

For the waveform generated from the program below, find: 1) the port and port bit, 2) frequency, and 3) duty cycle. Fosc = 8 MHz

```
.ORG 0
LDI R20, HIGH(RAMEND)
OUT SPH, R20
LDI R20, LOW(RAMEND)
OUT SPL, R20
LDI R16, 0X20
SBI DDRB, 5
LDI R18, -150
START:  SBI PORTB, 5
        OUT TCNT0, R18
        RCALL DELAY
        OUT TCNT0, R18
        RCALL DELAY
        CBI PORTB, 5
        OUT TCNT0, R18
        RCALL DELAY
        RJMP START

; TIMER DELAY
DELAY:  LDI R19, 0X02
        OUT TCCR0, R19
AGAIN:  IN R24, TIFR
        SBRS R24, TOV0
        RJMP AGAIN
        LDI R19, 0X00
        OUT TCCR0, R19
        LDI R19, 1<<TOV0
        OUT TIFR, R19
        RET
```

00000010

D D D

T = 450µs

$f = \frac{1}{T}$

prescalar

$t_{Delay} = 150\left(8\left(\frac{1}{8\times10^6}\right)\right)$

$T = \frac{1}{f_{osc}}$

66% duty cycle

= 150µs